

Sentiment Analysis using Machine Learning

Task 4 – Sentiment Analysis

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Track: Data Analytics

Objective

The objective of this project was to develop a Machine Learning model capable of automatically classifying Amazon customer reviews into Positive, Negative, and Neutral sentiments. The project focused on understanding Natural Language Processing (NLP) concepts while implementing an end-to-end sentiment analysis pipeline using Python and Scikit-learn.

Dataset Description

The project utilized the Amazon Reviews dataset obtained from Kaggle. The dataset contains customer reviews, review summaries, product information, helpfulness scores, and review ratings. The review ratings were converted into three sentiment classes:

- Positive
- Neutral
- Negative

Before model development, exploratory analysis was performed to inspect missing values and understand the class distribution. The dataset contained a significantly larger number of Positive reviews, making it an imbalanced classification problem.

Methodology

The project followed a systematic Natural Language Processing workflow.

The raw review text was first pre-processed by converting all text to lowercase, removing punctuation marks, tokenizing the sentences into words, eliminating English stopwords, and applying lemmatization to reduce words to their base form. These preprocessing steps reduced noise in the textual data and improved feature quality.

Since Machine Learning algorithms cannot directly process text, the cleaned reviews were converted into numerical feature vectors using the TF-IDF (Term Frequency–Inverse Document Frequency) Vectorizer. TF-IDF assigns higher importance to meaningful words while reducing the influence of commonly occurring words.

The dataset was divided into training and testing subsets using an 80:20 split with stratified sampling to preserve the original sentiment distribution.

Two Machine Learning algorithms were trained:

- Multinomial Naive Bayes
 - Logistic Regression
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Model Evaluation

Both models were evaluated using multiple performance metrics including Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

The Naive Bayes classifier achieved an overall accuracy of **79.20%** but struggled to classify Negative and Neutral reviews due to the highly imbalanced nature of the dataset. The model showed a tendency to predict Positive reviews for the majority of samples.

The Logistic Regression model significantly improved classification performance and achieved an accuracy of **87.69%**. It demonstrated better Precision, Recall, and F1-score across all sentiment classes and reduced the number of misclassified reviews considerably.

To better understand model performance, confusion matrices and comparison charts were generated. Additionally, WordCloud visualizations were created for each sentiment class to identify frequently occurring words.

Error analysis was also performed by examining misclassified reviews. Many incorrect predictions involved ambiguous wording, mixed opinions, or subtle sentiment expressions that were challenging for the model to interpret.

Results

Model	Accuracy
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Naive Bayes	79.20%
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Logistic Regression	87.69%
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Logistic Regression consistently outperformed Naive Bayes and was selected as the best-performing model for this project.

Conclusion

This project successfully demonstrated a complete sentiment analysis pipeline using Natural Language Processing and Machine Learning techniques. It provided practical exposure to text preprocessing, TF-IDF vectorization, supervised classification, model evaluation, visualization, and error analysis.

Among the two classifiers, Logistic Regression achieved superior performance and proved to be more effective for TF-IDF-based sentiment classification. Such sentiment analysis systems have practical applications in customer feedback analysis, product review monitoring, social media opinion mining, brand reputation management, and business intelligence.

The project strengthened the understanding of fundamental NLP concepts and demonstrated how Machine Learning can transform unstructured textual data into meaningful business insights.